

WEEK-8 Graded assignment

NAME :- Shahru Rohit Kumarh

Roll No :- 20F3100145

Q.1] $R_C = 10\text{K}\Omega$
 $R_E = 10\text{K}\Omega$
 $I_C = 0.5\text{mA}$
 $V_T = 25\text{mV}$

$$A_V = \frac{V_o}{V_i}$$

$$g_m = \frac{I_C}{V_T} = \frac{0.5 \times 10^{-3}}{25 \times 10^{-3}} = 0.02\text{ A/V}$$

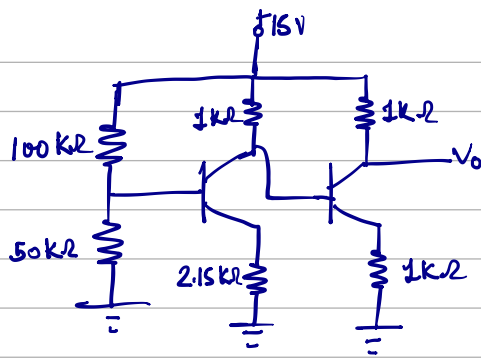
$$r_e = \frac{1}{g_m} = \frac{1}{0.02} = 50\Omega$$

$$\therefore \text{now } R_E = r_e // R_E = 5\text{K}\Omega$$

$$A_V = -g_m R_{C\text{eff}}$$

$$A_V = -0.02 \times 5000$$
$$= -100$$

Q.2)
Soln)



$$\text{Base voltage for first transistor} = \frac{15}{150} \times 50$$

$$= 5V$$

$$I_{E1} = \frac{5 - 0.7}{2.15k\Omega} = 2mA$$

$$\therefore V_{O1} = 15 - 2 \times 10^{-3} \times 10^3 = 13V$$

$$\text{now } V_{B2} = 13V$$

$$\therefore I_{E2} = \frac{12.3}{1k}$$

$$\text{now } V_O = 15 - \frac{12.3}{1k\Omega} \times 1k\Omega$$

$$V_O = 2.7V$$

Q.3)

Soln:

$$V_B = 18 \cdot \frac{110}{91+110}$$

$$= 9.85V$$

$$V_E = V_B - V_{BE} = 9.85 - 0.7 = 9.15V$$

$$I_E = \frac{9.15}{0.51 \times 10^3}$$

$$\therefore I_C = \frac{\beta}{\beta+1} I_E$$

$$\therefore I_C = I_E \times \frac{75}{76}$$

$$= 17.7 \times 10^{-3} \text{ A}$$

Q.4]
Soln:-

$$I_{REF} = \frac{V_{BE}}{R}$$

$$= \frac{0.7}{1000}$$

$$= 0.7 \text{ mA}$$

$$\therefore \text{for Transistor 1} = I_{B1} = \frac{I_{C1}}{\beta}$$

$$\text{For transistor 2} = I_{B2} = \frac{I_{C2}}{\beta}$$

$$I_{REF} = I_{C1} + I_{B1} + I_{B2}$$

$$I_{REF} = I_0 + \frac{I_0}{\beta} + \frac{I_0}{\beta}$$

$$I_{REF} = I_0 \left(1 + \frac{2}{\beta} \right)$$

$$I_0 = \frac{I_{REF}}{1 + \frac{2}{\beta}}$$

$$I_0 = \frac{0.7}{1 + \frac{2}{50}}$$

$$I_0 = 0.637 \text{ mA}$$

Q.5)

Soln:- Since both bases are tied together

$$\therefore V_E(\text{left}) = V_E(\text{right})$$

Emitter voltages

$$V_{E_L} = -V_{EE} + I_E(\text{left}) \cdot R_{E_L}(\text{left})$$

$$= -V_{EE} + (1\text{mA})(3\text{k}\Omega)$$

$$= -V_{EE} + 3\text{V}$$

$$V_E(\text{right}) = -V_{EE} + I_O R_{E_R}$$

$$= -V_{EE} + I_O (1\text{k}\Omega)$$

$$-V_{EE} + 3 = -V_{EE} + I_O \times 1000$$

$$I_O = 3\text{mA}$$

Q.6)

Soln :-

$$I_{C_1} = 25\mu\text{A}$$

$$I_B = \frac{I_{C_1}}{\beta} = \frac{25}{25} = 1\mu\text{A}$$

$$I_{B_2} = \frac{I_{C_2}}{\beta}$$

$$\text{now } I_{C_1} = \beta \cdot I_{B_1}$$

$$I_{C_2} = \beta \cdot I_{B_2}$$

$$I_{B_1} = I_{B_2}$$

$$I_{B_1} = \frac{25}{25} = 1\mu\text{A}$$

$$I_{C_2} = \beta \cdot I_{B_2} = 25\mu\text{A}$$

Q.7]

Soln Current through $4.7 \text{ k}\Omega$ res is

$$I_C = \frac{10 - 5.3}{4.7 \text{ k}\Omega}$$

$$I_C = \frac{4.7}{4700} = 1.0 \text{ mA}$$

Voltage across Resistor R

$$V_R = V_{CE} - (-10)$$

$$= 5.3 + 10 \\ = 15.3 \text{ V}$$

$$R = \frac{V_R}{I}$$

$$= \frac{15.3}{1.0 \text{ mA}}$$

$$= 15.3 \text{ k}\Omega$$

$$R = 15.3 \text{ k}\Omega$$

Q.8)

Soln:-

$$C_{MRR} = \frac{A_d}{C}$$

$$\therefore C_{MRR} = \frac{2000}{0.2} = 10000$$

$$C_{MRR} = 10000$$

Q.9)

for Transistor 1

$$I_{RE} = I_{C1} + I_{B1} + I_{B2}$$

$$I_{RE} = I_C + 2I_B$$

$$= I_C \left(1 + \frac{2}{\beta} \right) \because I_B = \frac{I_C}{\beta}$$

$$V_{R_1} = V_{CC} - V_{BE}$$

$$= 10 - 0.7$$

$$= 9.3 \text{ V}$$

$$R_1 = \frac{V_{R_1}}{I_{RE}}$$

$$I_{RE} = 1 \text{ mA} \left(1 + \frac{2}{125} \right)$$

$$= 1 \text{ mA} \cdot (1.016)$$

$$= 1.016 \text{ mA}$$

$$R_1 = \frac{9.3}{1.016 \times 10^{-3}}$$

$$= 9.15 \text{ k}\Omega$$

Q.10)

Sol :- For $I_C = 10 \mu\text{A}$

$$I_{RE} = 10 \mu\text{A} \left(1 + \frac{2}{125} \right)$$

$$= 10.16 \mu\text{A}$$

$$R = \frac{9.3}{10.16 \times 10^{-6}}$$

$$= 915 \text{ k}\Omega$$